

## Manufacturing Process

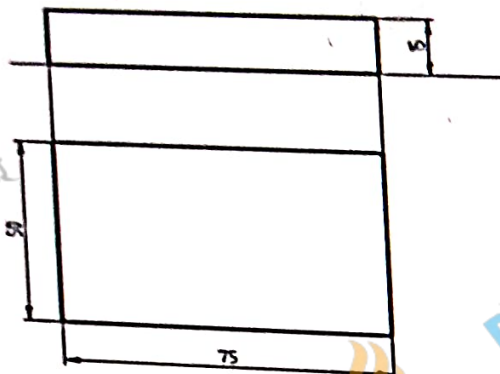
### Week-10

1. Explain the Development of Tray
2. Explain the Development of Transition piece
3. Explain the Development of Transition pieces

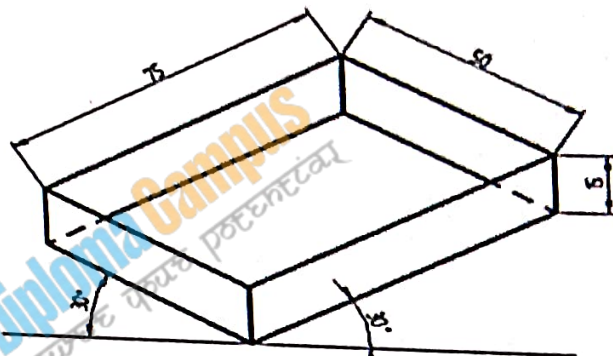
Experiment No.: 06

Date:

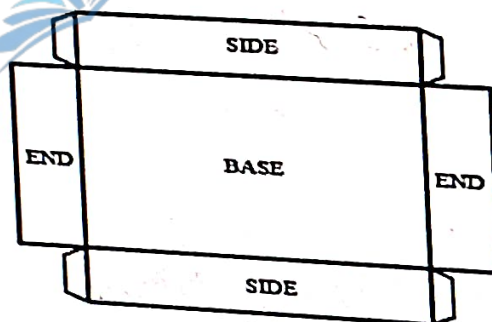
#### Development of Rectangular Tray



2D Drawing



3D Drawing (Isometric)



Development of Rectangular Tray

Sequence of operation or Procedure to prepare Trays in Sheet metal.

- Draw 2D/ 3D sketches as per scale.
- Draw the development considering all lines must be a "True Length"
- Construct the base of the Tray.
- Construct the two sides of the tray.
- Construct the two ends of the tray.



Visual Watermark

## **Manufacturing Process**

- Set 5mm extra allowance on side face for joining the corners by seaming.
- Notch the points to prevent bulging on seaming
- Cut the development shape on lines by using paper cutting scissor
- Place and fix the development on given G.P sheet by using Sticker.
- Punch two points on each bending line.
- Mark the boundary line by using Marking Pen.
- Remove the paper development join the two punch points by a scribe or marking pen.
- Shear the boundary lines by hand snip.
- Flattening the sheet by using mallet on flatter.
- Always bend the seaming line first and remaining lines to get the desired shape by forming.
- Finish the surface keeping on stake to shape.
- Solder the joint.
- Finish surface finishing and complete the Tray.



### Explain the Development of Transition piece

Transition Pieces are usually made to connect two different shaped objects (e.g. round pipes to square pipes) as often found in Ducting for Air Conditioning or Dust and Fume Extraction systems. The Development is constructed by assuming the surface to be made from a series of triangular surfaces laid side-by-side. This technique is known commonly as Triangulation and more specifically as Trilateration.

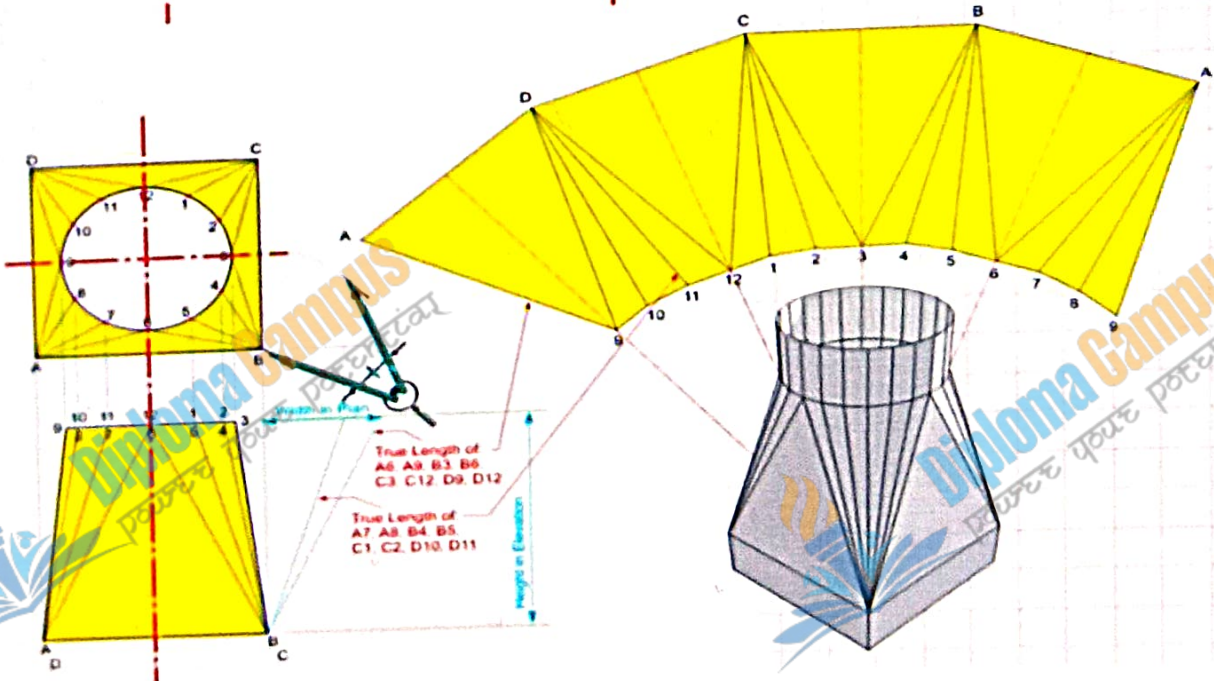
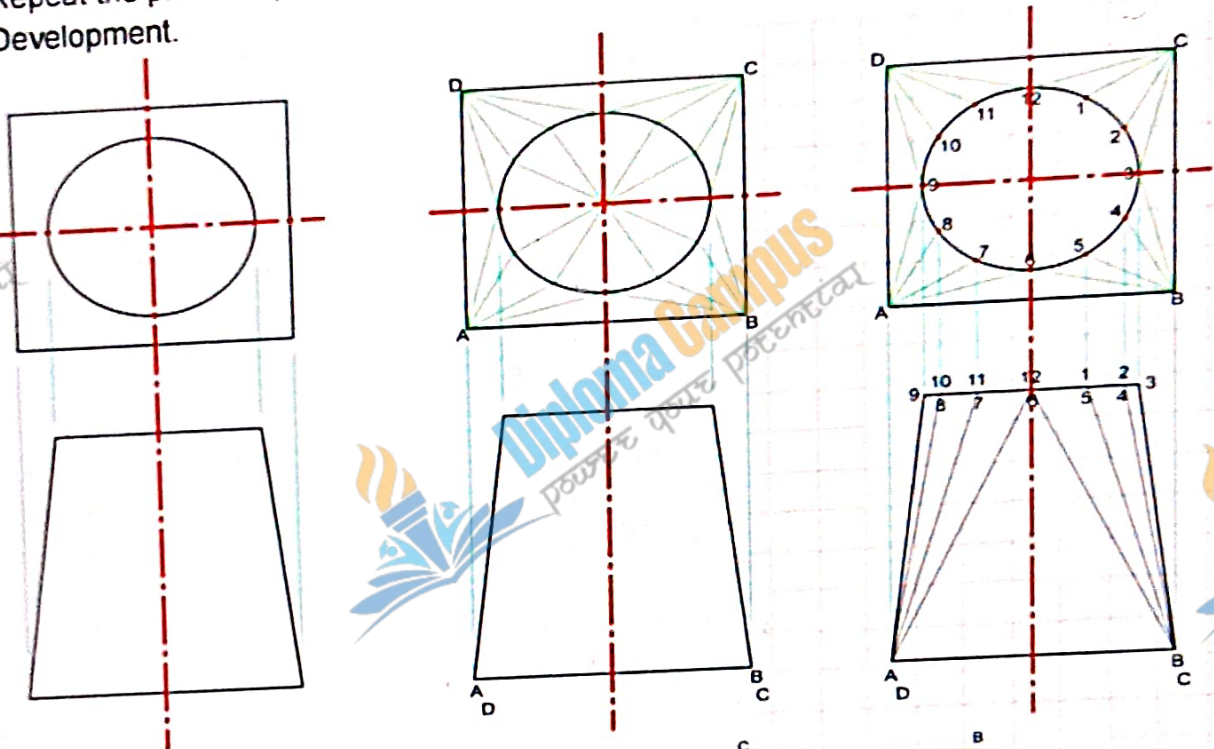
### Construction for a Square to Round Transition Piece

Consider the example of a Square to Round Transition Piece with both Axes directly in line with each other.

- Draw Plan and Elevation of the required Transition Piece,
- Divide the circle into, say, 12 equal points
- Join the corners of the square to the relevant points around the circle (Note : none of these are True Views because they are angled either away from or towards you),
- Number all points carefully,

## Manufacturing Process

- Use Dividers to transfer the lengths of the joining lines on to a horizontal line as shown (Note : in this case there are only two lines different in length).
- Using the Transition Piece elevation as height and the joining line lengths in Plan as base, draw the hypotenuse to find True Lengths.
- Work around the Plan View and use True Lengths to construct triangles as follows:
- Draw a line to the length of side A-D
- Set Compasses to the True Lengths of A-9 and D-9 and strike arcs from each end of A-D to create the Isosceles Triangle A-D-9,
- Similarly strike arcs equal to True Length D-10 and circle division 9-10 to find point 10.
- Repeat the process, paying careful attention to the point numbering, to construct the full Development.



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